Education of Nations: Against the Wealth of Nations

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In memory of Richard A. Easterlin (1926–2024), who asked the right question.

Abstract

Development theory since Adam Smith has treated education as one input among many — sometimes crediting income growth, sometimes direct state provision. Both are wrong about cause. Education is the cause of human development: the only input without which no other mechanism operates, and the only one that survives its own removal test. Remove income, and education still works: Bangladesh reached longer lives and fewer children per woman than the 1960 United States — at $1,159 per capita. Remove provision, and education still works: South Korea and Taiwan achieved development with minimal welfare states. Remove education, and nothing works. By 2022, 154 countries representing 80% of the world's population have crossed both thresholds. The remaining 20% is concentrated in sub-Saharan Africa, plus Pakistan, Afghanistan, and Yemen as the largest exceptions outside Africa.

In 187-country panel data (WCDE v3, 1960–2015) with country fixed effects and a 25-year generational lag, the decisive evidence is threefold. First, education predicts future life expectancy while GDP does not: ed-

ucation explains 47% of within-country variation in life expectancy one generation forward, GDP explains 17% — but when GDP is residualized against education (removing education's contribution to income), GDP's independent predictive power falls to near zero ($R^2 < 0.025$, $p > 0.1$) on every outcome tested: life expectancy, fertility, child mortality, and the next generation's education. Second, education's predictive power persists across three generational depths while income's collapses within one generation (Figure 3). Third, at no point in any country's trajectory does educational attainment decline from one generation to the next. Parental Transmission (PT) makes the gains irreversible; agency — the household decisions of educated adults — produces the development outcomes ~25 years later.

Sen's distinction between growth-mediated and support-led security (Drèze & Sen, 1989) — the most influential framework in development policy — classifies pathways by whether welfare arrives through income or through state provision, implicitly treating both as independent mechanisms. Neither is. Both are downstream of education. The two categories are the same mechanism producing different surface appearances depending on expansion speed and the presence of functioning markets. Every canonical case crossed the development threshold at the interval that prior education investment predicts, not from when provision programmes were introduced. These findings extend Easterlin (1981) and Lutz & Kebede (2018) to within-country identification of the generational mechanism.

Keywords: education; generational transmission; human development; Preston Curve; PT; agency; income-led development; provision-led development; education policy

1. Introduction

The error has a specific origin. In 1776, Adam Smith observed the pin factory and built a theory of wealth from the division of labour. He never asked where the educated workforce came from that made division of labour possible. Smith was born in Kirkcaldy in 1723, twenty-seven years after Scotland's Education Act of 1696 codified John Knox's programme of universal parish schooling into statute. The workforce he observed was the product of 150 years of deliberate educational investment — the grandchildren of Knox's first generation of pupils. Smith took educated labour as given. The tradition he founded inherited that assumption, and development economics has been working backwards from it ever since.

Smith's error was not a personal failure of observation. It was structurally invisible. For the entire pre-industrial period — roughly ten thousand years from agriculture to the steam engine — the world was Malthusian: total wealth was approximately fixed, and any gain by one group came at the expense of another. In a zero-sum world, hierarchy suppresses mass education rationally — the literate peasant is a danger, not an asset, when there is no new wealth to create. The equilibrium was self-reinforcing: no surplus to fund mass schooling, no mass schooling to generate surplus. What broke it was a contingent alignment. John Knox, for purely religious reasons — every person must read the Bible — broke the knowledge monopoly in 1560, two centuries before the steam engine. Scotland's education did not make Scotland rich; the zero-sum constraint held. But when fossil fuels arrived, they encountered a population already educated enough to deploy the new energy productively. Smith stood at exactly this inflection point — the first generation in human history where educated labour produced visible, compounding wealth — and mistook the output (division of labour, productivity) for the cause. The countries that remained in inertia after 1800 were not uniformly missing energy — many possessed coal and oil deposits. What they were missing was an educated workforce to convert energy into development. Japan, with negligible domestic fossil fuels, industrialised within a generation of the Meiji education reforms; Nigeria, sitting on one of the world's largest oil reserves, did not.

Easterlin (1981) was the first to ask the question Smith did not: "Why Isn't the Whole World Developed?" His answer — the binding constraint was the

absence of mass schooling systems, whose historical distribution tracked the Protestant Reformation's literacy imperative — pointed directly at the omission. Lutz & Kebede (2018) showed that the Preston Curve, the canonical relationship between income and life expectancy, is tighter when education replaces income on the x-axis: what looked like "wealth buys health" was education proxying through income all along. Both identified the correction. Neither stated the conclusion it implies.

This paper states it. Education is the cause of human development. Income is downstream. Direct provision is downstream. In *Hunger and Public Action* (Drèze & Sen, 1989), Sen codified the distinction that became the most influential framework in development policy for four decades: some countries secure welfare through economic growth (growth-mediated security); others through direct state provision regardless of income (support-led security). Kerala, Sri Lanka, and China were the paradigm cases of the second type — populations with welfare outcomes far above what their incomes predicted. This taxonomy is the most consequential expression of the assumption Smith introduced — that education can be taken as given.

My argument is not that Sen overlooked education, but that his taxonomy — by classifying pathways as either growth-mediated or support-led — implicitly treats both income and provision as independent mechanisms. They are not. Both are downstream of education. Sen's two categories are the same mechanism — education — producing two different surface appearances depending on the speed of educational expansion and the presence of functioning markets. Where educational expansion is fast and markets function, the mechanism produces rapid visible income growth; that income growth gets credited as the cause. Where expansion is slow and markets are absent, income growth is invisible; state services are the only observable welfare activity and they get credited instead. The taxonomy describes how resources flow through different institutional channels (markets vs. state services), not the underlying cause of development — but development policy read it as causation, and allocated accordingly. International aid now favours health over education at 2–3× (OECD DAC, 2022), and the SDGs rank education fourth — behind hunger, which the UN's own Human Development Index does not include.

Education is the cause — the only input that cannot be removed by economic collapse, state failure, or external shock. Income can be removed by economic crisis; provision can be removed by fiscal collapse; both happen routinely. Education cannot be removed because it is embodied in people, not stored in budgets or institutions. The removal tests (Section 4; Section 7) confirm this asymmetry across every available natural experiment. No other input survives its own removal.

2. Defining Development

I operationalize development concretely: a country is developed when it has simultaneously achieved a **total fertility rate (TFR) below 3.65** and **life expectancy > 69.8 years** — the 1960 United States values (World Bank WDI).

This definition is deliberate and theoretically motivated on three grounds.

First, 1960 is the natural baseline for the post-colonial world. Seventeen African countries gained independence in 1960 alone; the decade marks the inflection point at which most of the developing world began its independent trajectory. The development question is precisely: from the moment of decolonization, which countries achieved human welfare outcomes that surpassed the global hegemon? That is a theoretically precise question, not an arbitrary one.

Second, the USA in 1960 was the unambiguous global hegemon — the world's largest economy and dominant military power for fifteen years. A country that crosses both thresholds — lower TFR, higher life expectancy — has surpassed the richest, most powerful nation on earth on the metrics that matter for human welfare. The bar is meaningful.

Third, the USA in 1960 was itself still mid-demographic transition. Its TFR of 3.65 is the baby boom peak — the country had not yet completed the fertility transition (it crossed in 1972). Crossing below that threshold means going further than the hegemon had gone, not meeting a low standard.

The definition measures what matters. TFR and LE are chosen because they have intrinsic value: how long people live and whether women control their own fertility are ends in themselves, not proxy variables selected to flatter a

model. They require neither high income nor a particular type of government to achieve, making them appropriate for cross-regime, cross-income comparison. That education predicts them is an empirical finding, not a design choice. The circularity objection — that I chose outcomes education predicts — is addressed by Table A4: crossing dates under three threshold specifications shift by 6–35 years but the ordering and the PT-prediction alignment hold. The shift range is itself diagnostic: countries with high expansion rates cross any threshold within 6–7 years. GDP is excluded by design: measuring development through income would beg the question against an education-primary account.

The countries that had already crossed both thresholds by 1960 were those with the longest prior histories of mass education: Scandinavia, the Netherlands, France, Switzerland, the United Kingdom, Australia, and New Zealand. Japan — despite a TFR of 2.0 — had life expectancy of 67.7, just below the threshold; it crossed in 1964. The development dating analysis is robust to alternative threshold specifications (Appendix Table A4).

By 2022, 154 countries have crossed both thresholds, representing 80% of the world's population. Figure 1 traces this accumulation over time.

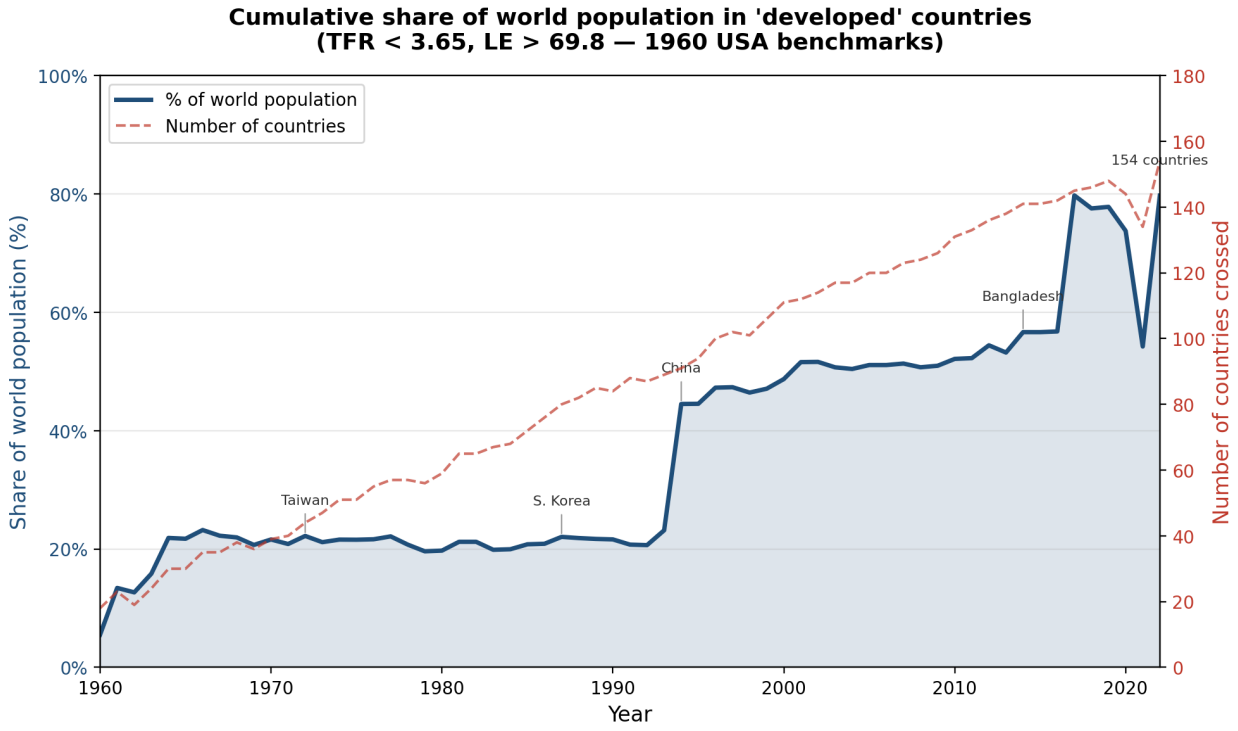

Figure 1: Cumulative share of world population crossing both development thresholds (TFR < 3.65, LE > 69.8), 1960–2022. Slow climb from 13% (1961) to ~20% (1993); vertical jump in 1994 when China crosses; 50% by 2001; 80% by 2017–2022. *(Source: World Bank WDI, WCDE v3. See* `scripts/fig_cumulative_developed.py`*)*

3. Theoretical Framework

3.1 Easterlin's Founding Argument

The education-first account of development has a precise starting point. Easterlin (1981), in "Why Isn't the Whole World Developed?", asked why modern economic growth had failed to diffuse uniformly given the post-war availability of production technology. His answer was specific: the binding constraint was the absence of mass schooling systems. Technology transfers when it encounters a literate, numerate population capable of deploying it. Without that population, it does not transfer, regardless of capital access or external assistance.

Easterlin located this empirically in the historical spread of education systems. The divergence in economic growth between early and late industrializers tracked the divergence in mass schooling — driven originally by the Protestant Reformation's insistence on personal scripture-reading, which produced mass literacy as a side-effect of religious practice across Northern Europe. The distribution of development in 1960 was substantially the distribution of schooling in 1860. Goldin & Katz (2008) provide the most thorough historical documentation of this transmission for the United States: the US "human capital century" of 1910–1940, driven by the high school movement, explains the majority of twentieth-century income growth — a within-country longitudinal result consistent with the cross-country generational mechanism estimated here.

3.2 Lutz and the Preston Curve

Lutz & Kebede (2018) take Easterlin's historical argument into the data. They reinterpret the Preston Curve (Preston 1975) — the positive relationship between national income and life expectancy — by replacing income with education on the x-axis. Three findings emerge. First, the education–life expectancy relationship is tighter than the income–life expectancy relationship. Second, and more important, the upward shift of the curve over time — the fact that populations live longer today at the same income levels than in 1960 — disappears when education is on the x-axis (except at the lowest education level, explained below). The "wealth buys health" reading of the Preston Curve is

an education story that income was proxying for, because income and education are correlated but income is downstream of education in the causal chain (Lutz 2009).

Third, and directly relevant to the scope of educational investment: the education–life expectancy relationship shows no diminishing returns at higher education levels. Unlike income, which follows the classic concave Preston shape, the education curve remains approximately linear from primary through tertiary. Lutz attributes this to a cognitive mechanism: education changes the brain's capacity for planning, learning from mistakes, and health-oriented decision-making across the entire life course, and these gains accumulate at all levels of schooling. The implication is that the case for educational investment does not weaken as countries progress toward universal secondary; pushing further into upper secondary and tertiary produces continued life expectancy gains. This is consistent with the WCDE panel: among 70 countries with lower-secondary completion above 85% in 2010, college completion carries a within-group correlation of r=0.44 with life expectancy. Mean life expectancy rises from 73.5 years in the lowest college-completion quartile to 79.0 years in the highest — a 5.5-year gradient among countries that have already solved the lower-secondary problem (see `scripts/verify_college_le_gradient.py`).

Deaton (2013) attributes the curve's upward shift to global health technology diffusion independent of education. But health technology was created by educated populations and its differential uptake tracks education; Deaton's account does not identify an independent mechanism (Section 5.3). If health technology diffusion were an independent channel, countries with comparable access to the same global health infrastructure would show comparable outcomes regardless of education level. They do not. Vaccines, antibiotics, and oral rehydration protocols are cheap, well-understood, and globally available — the technology has diffused. But a vaccine in a warehouse is not a vaccinated child. Between the existence of a technology and its uptake sits every household decision this paper describes: whether a mother recognises symptoms, follows the protocol, spaces pregnancies, takes her child to the clinic. The technology is globally available; the capacity to convert it into behaviour is locally determined by education. The existence of diffusion *without* uniform outcomes is itself evidence that education is the operative variable.

Lutz & Kebede (2018) document one further finding: life expectancy at the lowest education level in 2010 *exceeds* the 1970 baseline for that same level. This is the only level at which a genuine upward Preston Curve shift remains when education replaces income on the x-axis. The residual is education-mediated provision operating through international institutions: educated actors produced vaccines, antibiotics, and oral rehydration protocols; educated supply chains distributed them to the populations with the lowest educational bases. Sen's "support-led security" in its MDG-era form is not evidence that provision works without education. It is evidence that a sufficiently educated global class can deliver welfare externally to populations still building their own educational base — temporarily and without accumulating.

3.3 The Generational Transmission Mechanism: PT and Agency

The mechanism connecting these observations is generational transmission. **PT (Parental Transmission)** is cultural transmission from parent to child. The channel is biological, but what travels through it is cultural. The paper measures PT through education completion because that is the measurable proxy; what parents actually transmit is broader — norms, aspirations, health behaviours, the decision to send the next child to school. PT is the base unit. It transmits across generations: the grandchild's attainment reflects the grandparent's through two sequential cycles (~50 years), the great-grandchild's through three (~75 years).

PT explains persistence — why educational gains never depreciate. But persistence alone does not produce development outcomes. **Agency** does: the categorical reorganisation of the brain that formal education produces (Section 5.2) expresses itself across a lifetime of household decisions — fertility, health, investment, and children's schooling. The ~25-year lag between educational investment and development outcomes is the interval for educated children to become adults exercising these decisions. Taiwan's development crossing arrived ~20 years after educational expansion began: one generation of agency. Kerala's arrived at ~65 years: three generations of slow accumulation before enough of the population was educated for agency to produce population-level development outcomes. The variation in lag length follows

from biology, not assumption: a woman completing lower secondary at age 15–18 has children reaching school age 20–30 years later. Educational expansion has two parts. PT provides durability: it transmits what parents already have. State investment provides reach: it brings first-generation learners into the system and lifts the level beyond what parents alone would transmit. Where the state invests, both channels operate together — parents transmit what they have while the state reaches children whose parents have nothing (Section 6.1). Where the state does not invest, PT alone preserves what the parent already has — nearly perfectly, but nothing more.

The variation across countries is not a taxonomy of distinct mechanisms but three speeds of the same one. **PT alone** is the baseline: a parent transmits what they have, the next generation does the same. Without state investment, this is the only channel — century-scale, as in pre-modern Europe, which took 100–150 years to complete the S-curve. **Competing priority**: the state builds schools, but education competes with health, nutrition, security, and governance for funding and political attention. Multi-generational, as in India and most post-colonial countries. **Singular priority**: the state makes education the unconditional national focus, building ahead of need. One-generation crossing — Korea, Singapore, Cuba, Meiji Japan. All three are education-mediated. What varies is how the state allocates above the PT baseline: dispersed or concentrated.

PT's durability is not a statistical curiosity — it reflects a biological channel. Humans have plastic brains, but plasticity alone explains nothing: a plastic brain in isolation learns nothing useful. What delivers knowledge to the plastic brain is the human juvenile dependency period — roughly 18 years, compared to 8 for chimpanzees, the longest in any species (Konner 2010). A dedicated learner embedded with a dedicated teacher for nearly two decades: long enough for cumulative cultural knowledge to be absorbed deeply enough to become the learner's own baseline. The learner then becomes the teacher. Plasticity is the hardware; the extended dependency is the protocol; the parent is the carrier (Boyd & Richerson 1985). The channel is biological in the sense that it requires the species-specific parent-child dependency period, not in the sense of genetic transmission — what travels through it is cultural knowledge, not DNA.

This architecture explains three properties. Non-depreciation: 18 years of sustained transmission consolidates knowledge so deeply that it persists for life. Formal education is the most powerful payload ever delivered through this channel. Amplification: when the state builds schools, it loads additional knowledge onto the same channel — the $\beta > 1$ result (Section 6.1). Catastrophic reset: because the knowledge lives in the teacher, not the institution, destroying the educated population leaves the next generation's plastic brains with nothing to absorb. Cambodia (Section 7.1) is this test: schools rebuilt within a decade, progress stalled for a generation. Knox did not invent parental transmission; he institutionalised it, channelling formal schooling through a mechanism as old as the species.

3.4 From Action to Talk: How Education Reaches Beyond the Household

The PT mechanism operates through action — the household decisions of educated adults. But education's effects extend beyond the household, and the nature of that extension changes as the radius widens. Two things widen it simultaneously. First, education expands the boundary of who counts as "my people" — literacy connects you to people you have never met, shared knowledge creates shared identity, coalitional psychology (Tooby & Cosmides 2010) allows flexible group boundaries that education stretches from kin to region to nation. Second, education releases surplus: educated parents have fewer children and nearly all survive, freeing time, money, and attention that a parent with seven children and three surviving does not have. The arc widens (who you invest in) and the constraint loosens (what you can invest). Neither alone is sufficient. Surplus without expanded identity stays within the family — wealth hoarding, dynasty-building. Expanded identity without surplus is talk with no resources behind it.

At the **household**, education is action. The parent raises the child, recognises symptoms, spaces births, sends the next child to school. The 18-year biological channel guarantees delivery. There is no choice to disperse — you educate the child in front of you. PT fires automatically; gains compound into the next generation.

At the **kin and identity** radius, education is still action but identity-directed.

Siblings tutor younger siblings — strongest during early transition, high fertility, low parental base. Extended family members in multigenerational households (South Asia, sub-Saharan Africa, East Asia) transmit educational norms at timescales shorter than the nuclear PT interval. Diaspora invest back in home communities — educated Keralites in the Gulf funding schools in Thrissur, Indian professionals abroad channelling resources to home states. The boundary between kin and non-kin is blurred: Hamilton's kin-preferential investment shades into coalitional identity defined by caste, region, ethnicity, and nationality. The investment is concrete, directed, and still compounding.

Beyond identity, education becomes **talk**: advocacy, normative pressure, policy campaigns, global health delivery. The actor has no identity stake. The consequences are real — outcomes get measured, aid gets redirected, the MDGs and SDGs track progress — but diluted across every good cause simultaneously. This is where competing priorities emerge. At the household level there is no competition: you educate your child. At the identity level the investment is still directed. It is only when educated advocates operate beyond identity that they disperse across health AND nutrition AND governance AND education simultaneously — the competing-priority pathway (Section 3.3) that explains why most countries expand slowly. Global concern — educated actors delivering welfare technology to populations with no identity connection — is kin relaxation at its widest: the boundary has expanded to include strangers, and the surplus exists to act on it.

The critical distinction is durability. Investment at the household and kin radii compounds into the next generation through PT. Identity-level investment builds institutions that persist beyond any individual donor. But externally delivered non-education interventions at the global radius — vaccines, nutrition programmes, oral rehydration — are durable only as long as the institutions maintaining them remain engaged. Non-education intervention buys time; educational investment changes trajectory. This is also the mechanism behind the global educational expansion that year fixed effects absorb (Section 5.3): domestic demand, international advocacy, and normative pressure from educated populations all contribute to a global education signal whose components need not be disentangled for the causal claim to hold.

The in-group/out-group boundary is the same mechanism running in reverse.

Violence and exclusion toward out-groups is the primate default — the tight boundary that directs investment inward and hostility outward. In a zero-sum world, hierarchy suppresses mass education rationally: educating the out-group threatens the in-group's position. The surface vocabularies change across eras and cultures; the underlying structure does not. What education does is relax this boundary. As the in-group expands — through shared literacy, shared institutions, children in the same schools — the out-group shrinks. Mass education dissolves the boundary at scale: once everyone's children are in the system, the out-group category empties. The failure mode that remains is not exclusion but dilution — educated advocates dispersing across every good cause instead of concentrating on the one that compounds. The dilution is structurally difficult to correct. Educated people consistently rate education as important — but "important" means one of many important things, ranked alongside health, nutrition, governance, and security. They cannot categorise it as *fundamental* — the one input without which the others do not operate — because that would require seeing their own cognition from outside: recognising that the capacity to evaluate, compare, and advocate is itself the product of education. Uneducated people cannot see the value of formal education because they have not experienced it. The people who could identify education as fundamental cannot, because they are inside it; the people who would benefit most cannot, because they are outside it. The historical exceptions confirm the rule: Knox acted from religious conviction, Meiji Japan from existential military threat, Korea from Cold War survival. In every case where the decisive commitment was made, the motivation was not developmental — it came from outside the development frame entirely. No one broke through the invisibility by seeing education more clearly; they broke through by not looking at development at all.

The theory predicts that education's predictive power should decay smoothly across lag lengths, not in discrete 25-year pulses — because real populations have continuous age structure. People are of all ages at all times. Educational investment enters the population continuously as successive cohorts complete schooling, and agency — the household decisions that produce development outcomes — is exercised by adults across their entire lifetimes. Figure 3 (Section 6.2) confirms this: gradual decay across lags 0–100, the signature of a continuous process rather than discrete generational cycles overlapping.

3.5 Demographic Structure and Resilience

The accumulation has a demographic structure that varies with the level of completion. Fertility declines continuously as female secondary completion rises. The biological mechanism is a shift in reproductive strategy, not an override of the reproductive drive: in high-mortality environments, the optimal strategy is quantity — have many children because some will die. Education changes both inputs to this calculus. Child survival rises (fewer births needed to achieve the same number of surviving children), and the planning horizon extends — a literate parent can see that investing more in fewer children produces better outcomes than hedging with numbers. The drive to maximise offspring success is unchanged; what changes is the strategy that serves it. Caldwell (1982) identified the reversal of intergenerational wealth flows — from children supporting parents to parents investing in children — as the trigger for fertility decline. But the steepest fertility decline occurs at primary education (R^2=0.65, Section 6.3), where women are barely literate and not calculating investment returns or wealth flows. What primary education gives is agency over preferences that already exist — the capacity to read a contraceptive instruction, understand a health protocol, act on a desire not to be pregnant continuously. Caldwell's wealth flow reversal is at best a secondary/tertiary phenomenon; the first and largest fertility decline happens before anyone is doing his calculus. Caldwell also cannot explain why fertility continues falling past the point where wealth constraints disappear: rich uneducated populations (Gulf states pre-education expansion) had high fertility despite unlimited resources. The variable is education, not wealth flows. The steepest demographic headwinds are at the lowest completion levels, where total fertility rates are typically above 4, population growth is rapid, and the gains from each educated cohort are diluted across a larger and faster-growing next generation. As completion rises, fertility falls, child survival improves, and household resources per child increase. The educational investment does not accelerate at any particular threshold — Korea's data shows steady state-driven expansion from ~25% to over 90% with no inflection at any point (Figure 2). Sustained state investment is required all the way to near-universal completion. Critically, PT is a household mechanism. It operates through millions of individual family decisions — whether to send a daughter to secondary school, how many children to have, whether to vac-

cinate, how to interpret health information. States can create conditions for these decisions; they cannot make them.

3.6 How Education Produces Development

Nutrition and physical security are not preconditions for educational expansion — they are consequences of it. The asymmetry operates in two directions: (1) every successful educational expansion was made through poor nutrition and insecurity, not after resolving them; (2) destroying education causes multigenerational damage while destroying security does not.

First, the historical baseline. Humans have lived at the margins of subsistence nutrition for most of recorded history. This is not a recent problem to be solved before educational investment can begin — it is the starting condition from which every successful educational expansion was made. Korea in 1953 was a devastated post-war country with widespread malnutrition. Japan in 1872 was a poor agrarian nation operating at subsistence. Nepal in 1990 had GDP per capita of $423 and lower-secondary completion of just 27%. Nutrition improved *because* education expanded — through better agricultural knowledge, health behaviours, and the income gains that educated workforces generate. There is no historical example of a population that improved nutrition first and then expanded education. The causal arrow runs education → nutrition at the scale of national development, not the reverse.

Second, what destruction proves. Physical insecurity is not a development-stopping condition — conflict is episodic and localised, and its frequency has declined as education has spread. Sri Lanka's 26-year civil war did not break the educational base — conflict slowed the LE crossing by roughly 12 years (Section 7), but development arrived in 1993. Myanmar's continuous civil war since 1948 — the world's longest-running — did not prevent TFR falling from 5.9 (1960) to 2.3 (2015) and LE rising from 44.1 to 65.3, all at GDP below $1,200 (World Bank WDI). Myanmar has crossed the TFR threshold but not the LE threshold (65.3 vs 69.8). The asymmetry is diagnostic: TFR responds to primary education, which Myanmar achieved; LE requires lower secondary, which conflict delayed. War takes life out of the individual's hands — violence kills in ways education cannot fully mediate — but fertility decisions remain in the individual's hands even during conflict. Across eight major con-

flict episodes (Myanmar, Sri Lanka, Bosnia, Cambodia, Rwanda, Colombia, Afghanistan), TFR never reversed during war, while LE dipped in five of the eight. Sri Lanka shows the same pattern: TFR crossed during the civil war (1983–2009), but LE stalled for two decades before resuming its climb after the war ended.

Only the targeted destruction of the educated population itself — Cambodia (Section 7.1) — resets the generational clock. Security loss is a disturbance the mechanism runs through; education loss takes a generation to repair because the parental baseline cannot be rebuilt from buildings alone. The contrast with Uganda (Section 4) confirms: the binding constraint is the educational foundation, not provision.

4. Causal Identification: The Bad Control Problem and Natural Experiments

Every model in this literature that controls for GDP commits what Pearl (2009) calls the bad control error (see also Angrist & Pischke 2009, ch. 3): conditioning on a variable that is itself caused by the variable of interest.

Education predicts GDP one generation forward (FE β=+0.012 log-points per 1pp education gain; Table 2). In the pooled panel, GDP appears to predict education forward at similar strength — but this symmetry is an artifact of averaging across all education levels. Restricting to countries below 10% completion, education predicts future child education at R^2=0.594 while GDP predicts future child education at R^2=0.290 — a 2× asymmetry that widens to 3.4× at the 30% cutoff (Section 6.2). The causal argument rests on this asymmetry, on durability (Figure 3), and on the parental income channel collapsing in magnitude once parental education is controlled. Including current GDP as a control blocks part of the education signal through the income channel. The result is a biased downward estimate of education's true effect.

My primary analysis excludes GDP from the policy residual — the difference between actual and predicted education outcomes — for precisely this reason. The residual measures education delivered above the generational baseline, without crediting any achievement to the income that education itself generated.

Without an educational foundation, development gains reverse. Uganda in 1960 had identical life expectancy to India (both 45.6); by 1965, Uganda had pulled ahead (48.3 vs 45.6) — health investment without educational investment produced early gains. But those gains could not sustain themselves. By 1975, India had crossed Uganda; by 1984, India was 12 years ahead. By 1980, Uganda's life expectancy had fallen to 43.5 — below its 1960 level. The difference is not funding — India's health services were no better resourced. It is that India's educational expansion, while slow, had been building a parental base that continued to operate through household decisions. Uganda had no comparable educational foundation; without it, the health gains reversed.

China is the comparison case: it had an educational foundation; Uganda did not. China crossed the LE threshold (Section 7). Uganda has not arrived after 65 years.

A note on causal identification is warranted before presenting this evidence. The canonical approach to identifying education effects is instrumental variables: Duflo (2001) exploits variation in school construction across Indonesian districts to identify the causal effect of education on individual earnings. Duflo's design works because the instrument varies education exposure at short-run individual timescales — the relevant lag between schooling and earnings is five to fifteen years. At 25-year generational timescales across 187 countries, the identification strategy is necessarily different — and stronger. It rests on three identification structures. First, temporal ordering: parental cohort precedes child cohort by biological necessity, ruling out reverse causality at the individual level. Second, the lag-decay shape (Figure 3; Section 6.2): education and income produce structurally different decay profiles that autocorrelation cannot explain. Third, and most directly, a battery of natural experiments — cases where parental education was exogenously disrupted or where other development inputs were removed while education persisted — produce outcomes that pure autocorrelation cannot predict. The battery includes: Cambodia (population destruction → generational shadow); Uganda vs India (education destruction vs thin base → divergent recovery); Sri Lanka (civil war, PT unbroken, development on schedule); Myanmar (73 years of conflict, outcomes moved on schedule); China (barefoot doctors dismantled, LE continued rising); Korea vs Philippines (identical colonial base, divergent state commitment); and the Asian Financial Crisis (income wiped, education

untouched). The cases are developed in Sections 7 and 9. Cambodia — the most direct available test of PT — is presented in Section 7 alongside the other case studies, where the WCDE data it relies on has been introduced.

5. Data and Empirical Strategy

5.1 Data

Education data are from WCDE v3 (Lutz et al. 2021): lower secondary completion rates for the 20–24 age cohort, both sexes, which reflects completed education rather than enrolment. All education percentages in this paper refer to this measure unless otherwise noted. Coverage: 187 countries, 1950–2015 at five-year intervals; the core generational panel uses 1975–2015 (1,683 country-years); the residualization analysis (Section 6.3) extends to T=1960–1990 with education interpolated to annual values for precise threshold identification. Long-run analysis uses a 28-country subsample extending back to 1900, selected for data availability; the generational coefficient varies systematically with baseline education level (Section 6.1). GDP data: World Bank, constant 2017 USD, per capita, log-transformed. Life expectancy, TFR, and under-5 mortality: World Bank WDI. Parental education: each country's lower secondary completion lagged 25 years (T−25).

5.2 Completion as the Operative Variable

The quality objection — that completion measures quantity rather than quality (Hanushek & Woessmann 2008, 2015) — applies to a different question. Hanushek & Woessmann's dataset begins in 1960 and spans roughly 40 years; their test score measures do not exist before international assessments began in the 1960s. The educational expansions that explain Korea, Taiwan, Sri Lanka, and Kerala all precede their data. More fundamentally, quality follows quantity in the historical sequence: a literate, numerate population is the precondition for any test to be administered. WCDE completion is the appropriate measure for a 50–100 year generational mechanism; Hanushek & Woessmann's quality measures capture downstream returns to education already achieved.

The quality objection also misidentifies what PT transmits. Formal education — even at the lowest quality — represents a categorical discontinuity from oral cultural transmission. The neuroscience is unambiguous: literacy physically restructures the brain's visual processing pathways (Dehaene et al. 2010), and numeracy recruits parietal circuits unavailable to innumerate populations (Gordon 2004; Frank et al. 2008). These are not marginal improvements on existing capacities — they are new cognitive technologies that do not develop without schooling, regardless of intelligence or environment. A stunted child who learns to read undergoes the same categorical reorganisation as a well-nourished one; the stunting reduces efficiency but does not eliminate the discontinuity. The distance from no formal education to any formal education is categorical; the distance from a bad school to a good school is marginal by comparison. PT fires on this categorical threshold — the household decision to send the next generation to school, the health and fertility behaviours that change with any education, the aspirational shift that follows. The transmission is triggered by exposure, not by test scores. The endemic disease and malnutrition objections make the same error: they ask whether disease or hunger reduces cognitive performance, when the operative variable is whether the child was in school at all.

5.3 Empirical Strategy

Primary specification — country fixed effects regression:

$$E_{it} = \alpha_i + \beta_1 E_{i,t-25} + \varepsilon_{it}$$

where E_{it} is education completion in country i at time t, $E_{i,t-25}$ is the same country's completion 25 years earlier (parental cohort), and α_i absorbs all time-invariant country characteristics including institutions, geography, culture, and colonial history. The coefficient is identified entirely from within-country variation over time.

Comparison specifications:

$$E_{it} = \alpha_i + \beta_2 \ln Y_{it} + \varepsilon_{it}$$

$$E_{it} = \alpha_i + \beta_1 E_{i,t-25} + \beta_2 \ln Y_{it} + \varepsilon_{it}$$

For development outcomes, forward-lagged by one generation (~25 years) to eliminate reverse causality:

$$O_{i,t+25} = \alpha_i + \gamma_1 E_{it} + \gamma_2 O_{it} + \varepsilon_{it}$$

where O_{it} controls for initial outcome levels. O_{it} is not a bad control in the Pearl (2009) sense: it is caused by *prior* education ($E_{i,t-25}$, the parental cohort), not by E_{it} itself — the generational mechanism means E_{it} has not yet had time to affect O_{it}. Conditioning on initial outcomes is conservative; it absorbs prior trajectory and makes γ_1 a lower bound on education's true effect.

Choice of education variable. The main analysis uses aggregate (both-sexes) lower secondary completion. Female completion produces stronger R^2 on every outcome tested (Section 6.3). The aggregate specification is retained as the more conservative test: if education predicts development outcomes and GDP does not using the weaker predictor, the claim holds a fortiori with the stronger one.

Residualization strategy. The comparison between education and GDP as predictors is confounded by the fact that education drives GDP growth. Countries develop serially: education rises first, GDP follows a generation later. To isolate GDP's *independent* predictive power, I residualize log GDP against education using country FE — regressing log GDP(T) on education(T) within countries, and extracting the residual: the component of GDP that is *not* explained by education. This residual is then used to predict development outcomes at T+25. If the residualized GDP R^2 approaches zero, GDP has no independent effect — its apparent predictive power was education flowing through income. This two-step procedure (Frisch-Waugh-Lovell) is applied across three education levels (primary, lower secondary, upper secondary), four outcomes (life expectancy, TFR, child mortality, child education), and multiple entry-cohort/ceiling specifications (see `scripts/edu_vs_gdp_residualized.py`, `scripts/edu_vs_gdp_tfr_residualized.py`, `scripts/edu_vs_gdp_child_edu_residualized.py`).

Entry-cohort design. The cutoff-based sample definition (e.g., "countries be-

low 30% completion") selects observations, not countries. A country at 28% in one period and 35% in the next contributes one observation inside the cutoff and one outside. This loses the country's full development trajectory. I supplement the cutoff analysis with an entry-cohort design: for each threshold (10% to 90%, in 1% increments), identify the first year each country crosses that threshold, then include all observations from that point forward — but only while education remains below a ceiling (50%–90%). Education data are interpolated from WCDE 5-year intervals to annual values (linear) for precise threshold identification. This tracks countries through their development trajectory while excluding high-education observations that mechanically compress the education coefficient (see `scripts/edu_vs_gdp_entry_ceiling.py`).

On year fixed effects. Year FEs absorb part of the education signal, not a confounder. Adding year FE to the country FE specification dramatically reduces the parental coefficient (Table A1; Section 6.1). Education levels rose across nearly all countries simultaneously after 1975. If that global trend were an external confounder — something outside education driving all countries upward — then removing it would reveal the true, smaller effect. But the trend is itself education-driven: educated populations advocate for educational expansion globally (Section 3.4), and year FEs absorb this signal along with any genuine confounders. Section 6.1 quantifies the absorption.

6. Results

6.1 Education vs. GDP as Predictors of Attainment

Among countries where the expansion is still active — parental completion below 30%, where most of the remaining 20% of humanity sits — education R^2=0.70 vs GDP R^2=0.21. Education leads by 3.3×. GDP's R^2 barely changes across cutoffs (0.21–0.29); education's jumps from 0.54 to 0.70. GDP has no S-curve structure — its R^2 is flat across all cutoffs. The 30% cutoff is not selected for favourable results: education leads GDP at every cutoff from 10% to 90%, with ratios ranging from 1.4× to 3.3× (see `scripts/education_vs_gdp_by_cutoff.py`).

Table 1. Country fixed effects regressions: child lower secondary completion on parental education and log GDP per capita. Countries with parental completion below 30% (active expansion); 655 country-years, 111 countries.

Model	Parental edu β	Log GDP β	R^2 (within)
(1) FE: child ~ parent	**1.354*** **	—	**0.699**
(2) FE: child ~ log GDP	—	13.668***	0.211

*Standard errors clustered by country. *** p<0.001. Ceiling countries excluded; including them mechanically compresses both coefficients toward parity (Section 6.1). Education leads GDP at every cutoff from 10%–90% (`scripts/education_vs`*
Female completion produces stronger R^2 on every outcome; the aggregate is the more conservative specification. The gender gap in children's completion collapses universally as parental education rises; above 50% parental completion, it reverses (WCDE v3).

The decisive test is the shape across lags (Figure 3; Section 6.2).

Education levels rose across nearly all countries simultaneously after 1975. One objection is that this global trend inflates the parental coefficient. Year fixed effects test this.

Year fixed effects absorb education, not a confounder. When country and year fixed effects are both included, the parental coefficient among countries below 30% completion yields β=0.740 (t=5.5, R^2=0.345, n=790, 138 countries). Year FEs absorb roughly half the coefficient (from 1.354 to 0.740). The question is what that absorbed component represents. If the global trend were an external confounder — health technology diffusion, trade integration, or Cold War competition driving both education and development simultaneously — then removing it should reveal a weaker true effect, and the pre-trend period should produce a weaker coefficient still. The opposite is observed: the all-country panel (1900–2015), which includes decades before any common global trend, produces β=1.24 (with ceiling observations above 90% excluded; see below). The pre-trend coefficient is *stronger*, not weaker — consistent with year FEs absorbing part of the education signal rather than removing a confounder (Section 5.3). The remaining β=0.740 is entirely within-country, within-period variation — a country whose parents were more educated than other countries at the same education level in the same year still produces more educated children. On the pooled panel, the two-way FE collapses to β=0.080 (Table A1).

β exceeds unity at every cutoff — the autocorrelation test. Mere autocorrelation of a slow-moving stock produces $\beta \leq 1$ — each generation preserving what the prior had, nothing more. The data shows β far above unity. Across all countries in the WCDE with sufficient data (1900–2015), the naïve fixed-effects β equals 1.05 — but this is a misleading average. It includes observations where parental completion is already above 90%, where β mechanically approaches zero because there is no room to grow. Including these ceiling observations drags the average toward unity, creating the false impression that transmission merely preserves rather than amplifies. Excluding observations where parental completion exceeds 90% yields $\beta=1.24$ ($R^2=0.778$); below 50%, $\beta=1.82$ ($R^2=0.804$); below 20%, $\beta=2.83$ ($R^2=0.721$). At every cutoff, β exceeds unity — gains amplify across generations because the state adds reach on top of what households transmit. This is the decisive test against the autocorrelation objection: mere autocorrelation of a slow-moving stock produces $\beta \leq 1$, never $\beta = 2.83$ (see `scripts/beta_by_ceiling_cutoff.py`). The decline from $\beta=1.24$ to $\beta=0.49$ in the post-1975 panel reflects the S-curve structure: the post-1975 panel oversamples countries near ceiling.

The S-curve within countries. Within individual countries, β varies systematically with baseline education level (Figure 2). The United States shows $\beta=1.9$ when its baseline was 13% (early 20th century), declining to $\beta=0.08$ when its baseline reached 92%. Korea shows $\beta=6.5$ at 1% baseline, declining through $\beta=3.6$, $\beta=1.8$, and $\beta=0.2$ as its baseline rose to 59%. Taiwan follows a nearly identical trajectory ($\beta=5.1$ at 1.2%). The Philippines (Section 9) shows a flatter curve: $\beta=4.4$ at 1.5% declining more slowly to $\beta=0.4$ at 49%. The S-curve structure is robust across all countries with sufficient historical depth. Bangladesh shows $\beta\approx2$ throughout its observed trajectory (1–18% baseline), consistent with a country still in the steep part of the S-curve. India follows a similar pattern at $\beta\approx1.6$ through 28% baseline.

The S-curve confirms the theoretical framework (Section 3.3). $\beta > 1$ at low baselines is the empirical signature of state reach — the system expanding faster than household transmission alone. β approaching zero at high baselines is ceiling compression, not diminishing returns (distinct from Lutz & Kebede's cross-level finding that education's effect on life expectancy shows no diminishing returns from primary through tertiary). The speed at which β collapses measures how fast a country developed: Korea's β fell from 6.5

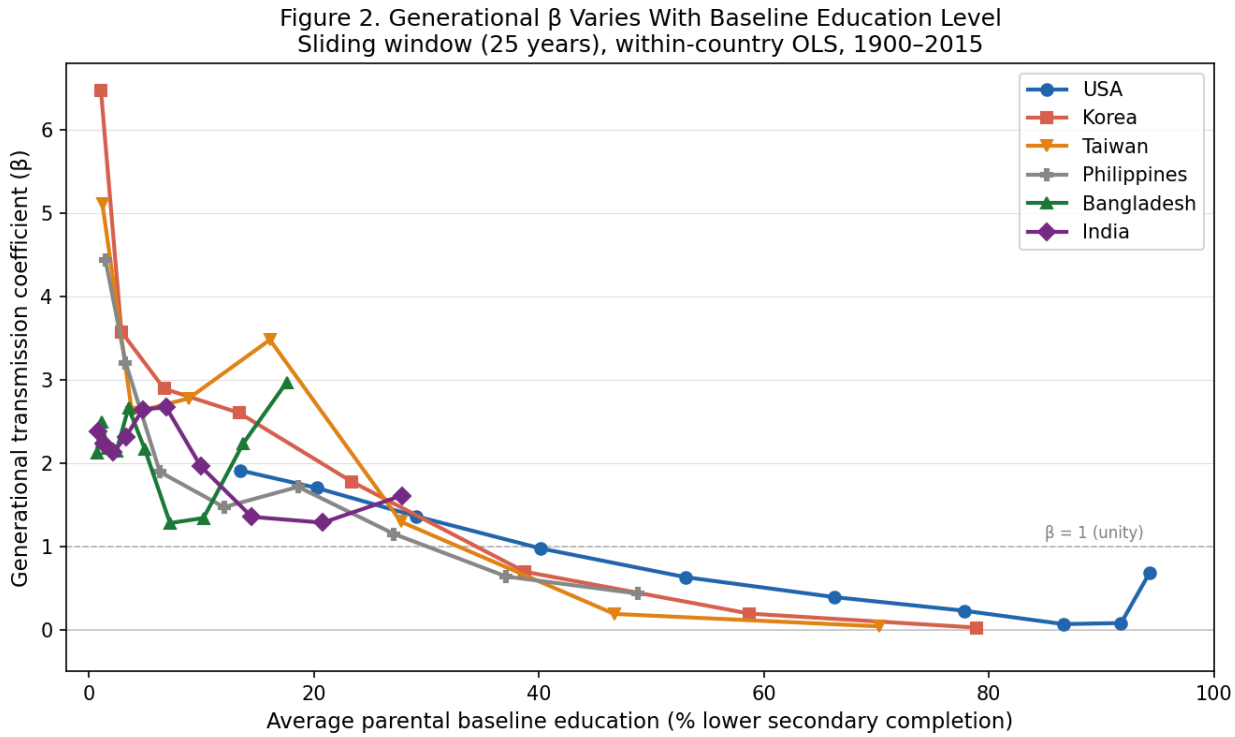

Figure 2: Generational β varies systematically with baseline education level. Each point represents a 25-year sliding window (6 child cohorts) within a single country, plotted against the average parental baseline for that window. $\beta > 1$ at low baselines reflects state reach; $\beta \to 0$ at high baselines reflects ceiling compression. *(Source: author's calculation, WCDE v3 long-run cohort reconstruction 1875–2015 — broader coverage than the 28-country regression panel in Table A3; Philippines historical data carries greater uncertainty for pre-1950 colonial-era observations. See* `analysis/scripts/fig_beta_vs_baseline.py`*)*

to near-zero in 60 years; the Philippines stretched the same range over 90 years; Bangladesh's β has held at ~2 for a century. Non-depreciation holds universally — the Cambodia test (Section 7.1).

Splitting the post-1975 panel by 1975 baseline confirms the S-curve structure: countries below 20% completion show β=1.585 (R^2=0.706, n=423, 47 countries), countries at 20–60% show β=0.713 (R^2=0.716, n=675, 75 countries), and countries above 60% show β=0.176 (R^2=0.442, n=585, 65 countries).

6.2 Education Predicts Development Outcomes 25 Years Forward

Table 2. Fixed effects regressions: development outcomes one generation forward (~25 years) on education, controlling for initial outcomes. Panel A reports education predicting development outcomes forward; Panel B reports the symmetric reverse regression (GDP predicting education forward) to demonstrate mutual endogeneity.

Panel A: Education(T) → development outcomes(T+25)

Outcome (one generation forward)	Education β	Initial outcome β	R^2 (within)
Log GDP per capita	+0.012***	+0.173***	0.354
Life expectancy (years)	+0.108***	+0.301***	0.384
Total fertility rate	−0.032***	+0.037*	0.367

The reverse regression tests whether GDP predicts future education with comparable strength — it does not once parental education is controlled.

Panel B: Log GDP(T) → education(T+25) — reverse regression

Outcome (one generation forward)	Log GDP β	Initial edu β	R^2 (within)
Lower secondary completion	+14.85***	—	0.272
Lower secondary completion	+3.780***	+0.485***	0.500

Country fixed effects, same panel. n=828 for GDP-education matched obs.

Each 1pp gain in lower secondary completion predicts, one generation forward (~25 years): 1.2% higher GDP per capita, 0.108 more years of life, and 0.032 fewer children per woman, identified from within-country variation after controlling for initial conditions.

When education and income are each used to predict future child education one generation forward, education dominates at every level. Restricting to countries below 10% completion — where confounding between the two variables is weakest — education(T) predicts child education(T+25) at R^2=0.594 while GDP(T) predicts child education(T+25) at R^2=0.290 — a 2× advantage. At the 30% cutoff, the ratio widens to 3.4× (R^2=0.701 vs 0.208, consistent with Table 1's 3.3×). At every cutoff from 10% to 90%, education predicts future child education better than GDP does (see `scripts/education_vs_gdp_by_cutoff.py`).

The asymmetry is sharper still when the outcome is life expectancy. Among countries below 30% completion, education(T) predicts life expectancy(T+25) at R^2=0.309; GDP(T) predicts life expectancy(T+25) at R^2=0.021 — education leads by 15× in the raw comparison. Among countries below 10% completion, education R^2=0.628 while GDP R^2=0.016 — effectively zero. Education's R^2 remains stable at approximately 0.30 across all cutoffs; GDP's R^2 never exceeds 0.03. But even this understates the asymmetry: as Section 6.3 shows, the small apparent GDP signal vanishes entirely once education's contribution to GDP is removed.

Figure 3 confirms this asymmetry across lag lengths. Measured across lag lengths from 0 to 100 years, education and income have structurally different decay profiles. Education's predictive power for life expectancy decays

slowly across three generational depths: R^2=0.562 at lag 0, R^2=0.364 at lag 25, R^2=0.171 at lag 50, R^2=0.085 at lag 75. Income's collapses within one generation — already fading by lag 20–25. Beyond lag 50, income data does not thin out — income itself was uniformly low. Every pre-literate subsistence society sat at approximately $400–600 per capita (constant 2017 USD). This is not missing data; it is the data: income had no cross-country variation because education had not yet created it. Backfilling pre-1960 income at $500 per capita for all countries does not rescue income's predictive power — income's R^2 still collapses to zero by lag 60, while education's remains positive at lag 100 (see `scripts/backfill_income_lag_test.py`). There is no hidden income signal at long lags because there was no income variation to predict with.

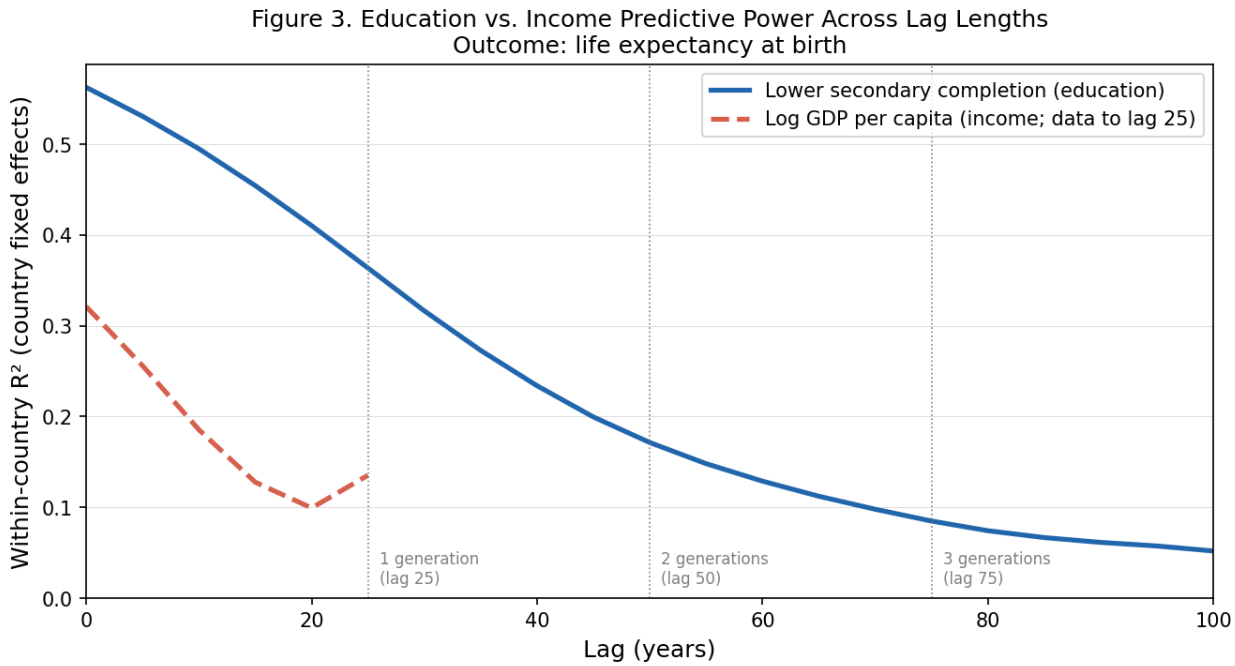

Figure 3: Education vs. income predictive power across lag lengths (within-country FE, outcome: life expectancy at birth). Within-country R^2 for lower secondary completion and log GDP per capita plotted against lag lengths from 0 to 100 years. Three generational horizons marked (lag 25, 50, 75). Income shown through lag 25 only; beyond that, income was uniformly low across pre-literate subsistence societies. *(Source: author's calculation, World Bank WDI and WCDE v3. See `analysis/scripts/fig_a1_lag_decay.py`)*

What lag 100 means in human terms: in countries that were still expanding education, a great-grandparent's completion predicts how long their great-grandchild will live — four generations later. No other development input has

a comparable reach.

6.3 GDP Has No Independent Effect

The comparison above overstates GDP's role. GDP(T) correlates with future outcomes partly because education(T) drives GDP growth — countries develop serially. To test whether GDP has any *independent* predictive power, I residualize GDP against education (Section 5.3) and use the residual — the part of GDP not explained by education — to predict each outcome one generation forward.

Table 2b. Residualized GDP analysis: education vs. raw GDP vs. GDP after removing education's contribution. Entry-cohort design (entry ≥ 10%, ceiling ≤ 90%), country fixed effects, lower secondary completion, T=1960–1990, lag=25 years.

Outcome (T+25)	Education R^2	Raw GDP R^2	Residualized GDP R^2	p-value	Education → GDP R^2
Life expectancy	0.472	0.165	**0.003**	0.56	0.417
Total fertility rate	0.478	0.175	**0.000**	0.98	0.417
Child education	0.524	0.303	**0.008**	0.31	0.417
Under-5 mortality	0.284	0.046	**0.023**	0.11	0.417

Clustered standard errors by country. Residualized GDP R^2 reports the within-R^2 of GDP residuals (orthogonal to education by construction) predicting each outcome. Education → GDP R^2 reports how much of within-country GDP variation is explained by education. See `scripts/edu_vs_gdp_residualized.py`, `scripts/edu_vs_gdp` `scripts/edu_vs_gdp_child_edu_residualized.py`.

The result is unambiguous: after removing education's contribution to GDP, income has no independent predictive power for any development outcome. The residualized GDP R^2 never exceeds 0.023, and the associated p-values

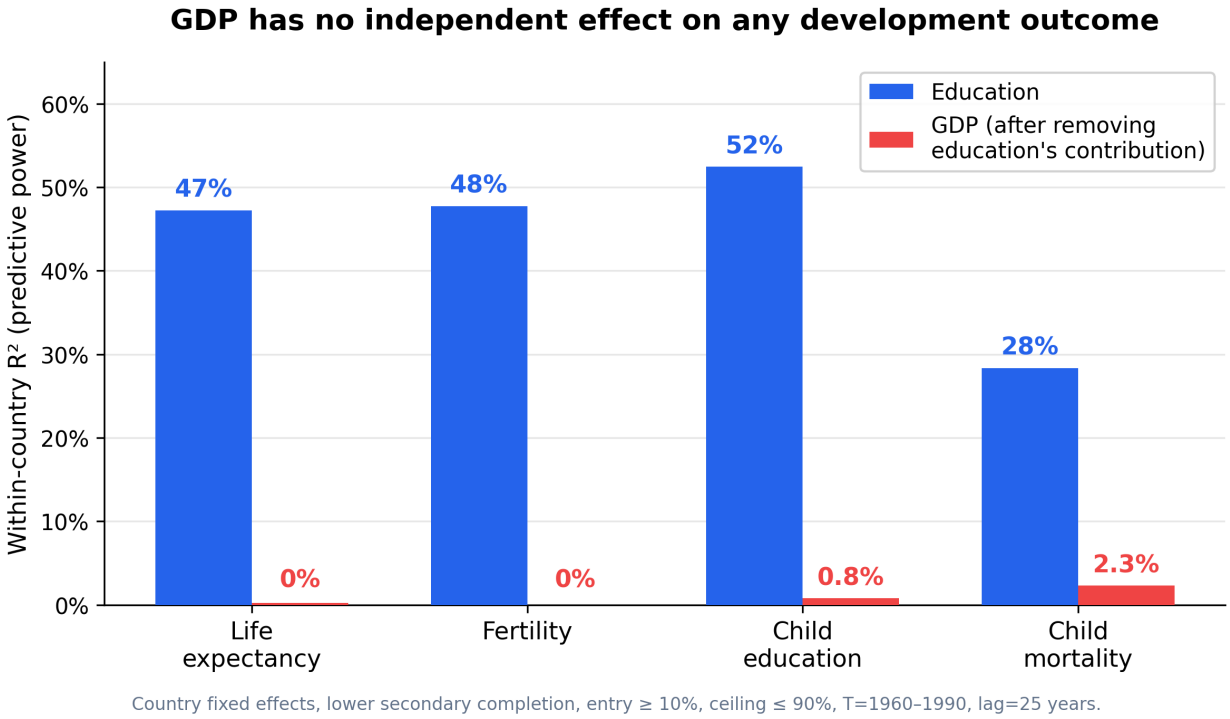

Figure 4: Education R^2 vs. residualized GDP R^2 across four development outcomes. Blue: within-country R^2 for education predicting each outcome one generation forward. Red: within-country R^2 for GDP *after removing education's contribution to GDP*. Entry-cohort design (entry $\geq$ 10%, ceiling $\leq$ 90%), country fixed effects, lower secondary completion, T=1960–1990, lag=25 years. *(Source: author's calculation. See* `scripts/fig_residualization.py`*)*

are all above 0.1 — statistically indistinguishable from zero. Education explains 42% of within-country GDP variation; the remaining 58% of GDP predicts nothing about the future. The weakest result — child mortality (p=0.11) — is driven entirely by post-2000 outcomes (pre-2000: R^2=0.008, p=0.33; post-2000: R^2=0.032, p=0.04), consistent with MDG-era health interventions producing a genuine upward shift in survival at the lowest education level — the one case where external provision temporarily substitutes for domestic education (Section 3.2). Splitting by parental education band sharpens this: among countries below 20% completion — where the remaining undeveloped population is concentrated — residualized GDP explains nothing in any period (pre-2000: R^2=0.003, p=0.70; post-2000: R^2=0.053, p=0.14). The post-2000 signal in the full sample comes entirely from countries at 30–50% completion, where the communication environment itself has shifted: educated producers, semi-educated audiences, and health information circulating through television, mobile phones, and social norms. At the bottom, where the policy decision matters, income buys nothing (see `scripts/u5mr_by_edu_level.py`).

This holds across education levels, but the *composition* of the effect varies in a way that reveals the mechanism. Primary education is the strongest predictor of fertility decline (R^2=0.65 vs 0.48 for lower secondary, 0.32 for upper secondary): the first educated generation — the categorical jump from no schooling to literacy — is where fertility falls hardest. By contrast, life expectancy gains accumulate with deeper education: upper secondary (R^2=0.48) predicts better than primary (R^2=0.41). This is precisely the structure that Lutz & Kebede (2018) identified when redrawing the Preston Curve with education on the x-axis — the relationship between education and health tightens at higher levels, while the fertility transition is triggered by the first level. The residualization confirms that income plays no independent role at any level: residualized GDP R^2 never exceeds 0.03 for life expectancy and is effectively zero for fertility, from primary through upper secondary.

The result is robust to lag specification: at lags of 15, 20, 25, and 30 years, residualized GDP R^2 remains below 0.02 for life expectancy and below 0.01 for fertility. A quadratic first stage (adding education^2) does not change the result: residualized GDP R^2 remains below 0.03. Bootstrapped 95% confidence intervals (1,000 country-cluster replications) confirm the gap: education R^2 [0.33, 0.59] vs residualized GDP R^2 [0.00, 0.04] — the intervals do not over-

lap. The Anderson-Hsiao IV estimator, which corrects for Nickell (1981) bias in short-T dynamic panels, produces coefficients within 3–10% of the standard FE estimates (see `scripts/robustness_tests.py`). Female education (lower secondary, female only) produces stronger R^2 than the aggregate on every outcome (0.531 vs 0.472 for life expectancy, 0.498 vs 0.478 for fertility); the main analysis uses the weaker aggregate specification throughout.

The parental income test is a textbook mediation decomposition. Regressing child education on parental log GDP (T−25) alone yields β=15.4 (R^2=0.293), but conditional on parental education the GDP coefficient drops by 72% to β=4.3 (p=0.04), adding only R^2=0.014 beyond education alone. Education's coefficient barely moves (β=0.475 from 0.553) while income's collapses in magnitude. Income transmits through education, not vice versa.

The Asian Financial Crisis (1997–98) provides the cleanest income-removal test. GDP collapsed abruptly and exogenously across five countries: Indonesia (−14.5%), Thailand (−8.8%), Malaysia (−9.6%), Philippines (−3.0%), and Korea (−6%). Lower secondary and upper secondary completion continued uninterrupted in all five countries. Indonesia gained 5.4pp of lower secondary completion over 1995–2000 while losing 14.5% of GDP. Thailand *accelerated* — lower secondary gained 13.4pp in the crisis window vs 10.0pp in the prior five years; upper secondary gained 9.8pp. College completion showed a cohort-level slowdown in Korea (gains fell from +4.5pp to +1.7pp per five years after 2000), but no country's college attainment declined — the generational mechanism (child exceeding parent) was never violated at any level. Income was removed at scale; education endured (see `scripts/asian_financial_crisis.py`).

The TFR result in Table 2 contradicts the standard competing explanation in demography. Initial fertility barely predicts subsequent fertility change once education is included (β=0.037). Demographic momentum does not drive the transition — educated women's decisions do.

6.4 Policy Over-Performers

Table 3. Countries delivering education above generational baseline, 2015 FE residuals. The residual measures how far a country exceeded its own predicted within-country trajectory — not a global average, but performance above each country's own historical trend.

Country	FE residual above baseline	GDP per capita (2015)
Maldives	+34.9 pp	$9,645
Cape Verde	+26.3 pp	$3,415
Bhutan	+26.1 pp	$2,954
Tunisia	+25.5 pp	$4,015
Nepal	+17.8 pp	$876
Viet Nam	+16.0 pp	$2,578
Bangladesh	+15.8 pp	$1,224
India	+14.1 pp	$1,584

Nepal at $876, Bangladesh at $1,224, Vietnam at $2,578: the premise that income is the prerequisite for educational over-performance is directly refuted. These countries made the decision to treat education as the primary investment and produced outcomes that outpaced both their demographic inheritance and their income.

The panel evidence establishes that education dominates income as a predictor across 187 countries. The following case studies trace the mechanism through the canonical cases in the development literature.

7. When Did Sen's Cases Actually Develop?

The education-mediated account predicts that development — as defined by the TFR and life expectancy thresholds — occurs one generation (~25 years) after sustained education investment, independent of whether state provision was present.

Table 4. Year of development (TFR < 3.65 AND life expectancy > 69.8) and relationship to prior education investment. Thresholds are 1960 United States values (World Bank WDI).

Country	Developed	TFR crossed	LE crossed	Expansion onset	Lag (yrs)	Generations
Taiwan	**~1970**	~1970	~1970	1950s	~20	**1**
S. Korea	**1987**	1975	1987	1953–65	~25	**1**

Country	Developed	TFR crossed	LE crossed	Expansion onset	Lag (yrs)	Generations
Cuba	**1974**	1972	1974	1961 (40% base)	~13	**1**
Bangladesh	**2014**	1995	2014	1990s	~24	**1**
Sri Lanka	**1993**	1981	1993	1940s–50s	~42	**2**
China	**1994**	1975	1994	1950s+Cultural Rev.	~42	**2**
Kerala†	**~1982**	~1973	~1981	Early 20thC	~65	**3**
Uganda	**—**	TFR 5.25	LE 63.8	None	—	—

Developed = year both thresholds crossed. Expansion onset = year or level at which sustained educational expansion began. Lag = years from onset to development crossing. Generations = number of ~25-year cycles. Cuba's high starting base (40%) and the 1961 reconstruction of losses from the post-revolution exodus shortened the lag below the typical ~25-year generation. † Kerala figures estimated from India Sample Registration System and census records; all other figures from World Bank WDI direct measurement.

Korea is the calibration case: 2.14 pp/yr, the fastest sustained expansion in the WCDE dataset. All other rows in Table 4 are independent tests. The expansion rate determines generations to crossing (Section 3.3): benchmark pace compresses it to one generation; India's 0.87 pp/yr spreads it across three.

The second parameter is structural disruption. Life expectancy gains flow from education through household health behaviours and the agency of educated populations who become the workforce that delivers health services — neither requires markets. What delays LE crossing is conflict that directly kills and displaces populations, or state dismantling of health services before the educated population can rebuild them. Sri Lanka's 12-year gap between TFR and LE crossing (1981→1993) maps onto the civil war's onset (1983) and its disruption of health services and physical security. China's 19-year TFR-to-LE gap (1975→1994) reflects Deng's dismantling of the barefoot doctor network

and China's larger absolute distance from the LE threshold. Both delays are disruption-mediated. Korea, Taiwan, Cuba, and Bangladesh had no comparable structural disruption during their transmission periods.

All three parameters — starting base, pace relative to the Korea benchmark, and presence of structural disruption — can be read from educational data and historical record before consulting Table 4's crossing dates. The predicted depths match the observed lag lengths: Korea-pace cases crossing in 20–34 years; half-Korea-pace cases crossing in 42–45 years absent disruption, longer with it; third-Korea-pace cases crossing in 60–70 years. The crossing dates are the test, not the inputs. Figure 3 confirms all three generational depths empirically.

Taiwan and Korea crossed earliest — both at benchmark pace, both state-driven — and both had functioning markets that let educated workers move into productive employment, generating income and life expectancy gains simultaneously. State-driven simultaneous expansion compressed development to 20–34 years against the 40–70 years seen in competing-priority cases.

Kerala crossed at ~1982 — the paradigm case of competing-priority expansion. The education had been building since the early twentieth century through social reform movements — gradual accumulation, demand-pulled, not state-driven at Korea pace. The micro-decision pathway fired (literate women made better health and fertility decisions) but there was no market-mediated economic translation of the kind Korea and Taiwan achieved. Kerala had extensive state provision — public health, food distribution, land reform. But contraceptive provision does not produce a fertility decline; it provides a tool. What determines fertility is intent, not access — and intent is education. Mali has contraceptive availability but low uptake — precisely because the educational base that generates demand is absent — and TFR above 6.

The TFR threshold was crossed by ~1974; the LE threshold followed ~8 years later. The contrast within India confirms this: during the Emergency (1975–77), forced sterilisation campaigns failed in the least-educated northern states and provoked a political backlash that brought down the government. Kerala needed no coercion; its educated women had already made the decision. State coercion either arrives after education has done the work, or fails where education has not.

Sri Lanka crossed in 1993. The TFR threshold was crossed in 1981, driven by education. Life expectancy climbed steadily to 69.0 by 1988, fell to 67.3 in 1989 during the civil war, and recovered to cross 69.8 in 1993. The war disrupted economic activity and life expectancy without breaking PT: the household transmission of educational aspiration continued through the conflict, and Sri Lanka's education metrics did not collapse. The delay is precisely the length of the war's worst period. This is the war disrupting economies, not disrupting education.

China crossed in 1994. Three corrections to the standard narrative are required here. First, the TFR threshold was crossed in 1975 — five years before the one-child policy (1980). The most famous population policy in human history was unnecessary: the threshold was already crossed before coercion began. Cai (2010) shows that China's fertility trajectory mirrors South Korea and Thailand's — both of which achieved equivalent TFRs without any compulsory policy. Miller et al. (2018) confirm that the Wan Xi Shao campaign accounts for less than 30% of China's 1970s fertility decline; the majority was driven by the same mechanism visible in every country at comparable education levels.

Second, what is universally called an educational catastrophe was in fact the largest educational expansion in Chinese history. The standard framing conflates the disruption of university education — affecting a small percentage of the population — with the overall educational trajectory. For rural China — approximately 80% of the population (National Bureau of Statistics 1982) — the Cultural Revolution era produced the largest lower secondary gains in China's entire WCDE 1870–2015 record: +10.6pp for the 1975 cohort and +15.0pp for the 1980 cohort. Both cohorts' secondary schooling fell within the Cultural Revolution years. Community schools (民办学校) brought secondary education to villages that had none (Pepper 1996; Unger 1982; Gao 2008). The base of the pyramid expanded dramatically while the apex was disrupted.

Third: the provision debate is misconceived. Sen's framework attributes China's LE gains to the barefoot doctor (赤脚医生) program. The cross-country data shows this debate is the wrong one to have. Countries at comparable lower secondary completion (~30%) in 1965 had life expectancies of 54–65 years, with a peer average of 57.4. This held regardless of whether they had

barefoot doctors, Cuban-style health campaigns, or neither. The LE outcome tracks education level, not the provision model. China's 54.4 was anomalously low for its education level, depressed by the Great Leap Forward famine (1959–1961). The +10-year gain to 64 by 1980 was nearly double the peer average gain of +6 years over the same period (Philippines +1.5, Peru +6.1, Panama +7.1, Vietnam +8.4) — China was recovering to the LE that its education level predicted. By 1980, at 62% lower secondary completion and 64 years LE, China had converged to exactly where its education-peer group sat.

The barefoot doctors and the community school teachers came from the same CR-era rural mobilisation. When Deng dismantled the barefoot doctor system after 1980, LE continued to rise — from approximately 64 to 69.8 by 1994 — because the educational baseline built into rural households did not leave with the doctors.

Drèze & Sen (1989, pp. 215–221) document the resulting health crises and rising rural medical costs (Gao 2008; Liu et al. 2003). But among all countries that crossed 45% lower secondary completion (n=108), China's LE gains at T+5, T+10, T+15, and T+20 exceeded the peer mean at every horizon — and the gap *widened* after the barefoot doctors were dismantled, not narrowed (see `scripts/china_provision_discontinuity.py`). Lower secondary completion rose from 30.9% (1965) to 62.0% (1980) to 75.0% (1990), tracking LE improvement across both eras.

China crossed the LE threshold in 1994 — later than Taiwan (~1970) and Korea (1987) — not because provision was dismantled, but because China started further from the threshold: Korea's 1965 LE was 55.9 (14 years below 69.8), China's 1965 LE was 54.4 (15 years below). At a roughly constant rate of improvement, the crossing dates follow mechanically from starting distance. And China started lower because the Great Leap Forward famine had depressed its baseline — the same event that Sen's framework treats as the starting point for a provision-led welfare success story.

Cuba crossed in 1974 — four years after Taiwan, 13 years after the 1961 literacy campaign. Cuba's educational base was substantial before the revolution: lower secondary completion was 40.3% in 1960. But the post-revolution exodus (1959–1962) removed much of the professional class — doctors, lawyers, engineers, teachers. The 1961 literacy campaign was Cuba's barefoot teacher

moment: 268,000 volunteers deployed to rebuild and expand the educational base (Prieto 1981) that emigration had hollowed out. This was not finishing a job — it was a decisive educational expansion, parallel to China's community schools (民办学校), replacing a professional elite that left with a mass household baseline that stayed. LE was 63.3 in 1960 and climbed steadily to 69.9 by 1974. Sen credits Cuba's health system; the evidence points to Cuba's teachers. That Cuba and Taiwan — one Soviet-aligned, one US-allied developmental authoritarian — crossed within four years of each other is the regime-independence test.

Bangladesh crossed in 2014 — the most important case in the table. Lower secondary completion was 11.4% in 1960 and GDP per capita was $1,159 in 2014. Bangladesh crossed both development thresholds through sustained state commitment to girls' education — primary and secondary expansion from the 1990s — confirming the universal pattern: as parental education rises, the gender gap in children's completion collapses (Table 1 note). The garment industry provided female economic participation that reinforced educational agency. IIASA's multilevel analysis confirms that increasing female education was the main driver of Bangladesh's fertility decline, operating both at the individual level and through community diffusion (Mamun & Bongaarts 2022). TFR decline tracks girls' secondary expansion, not contraceptive distribution. The LE and TFR gains came through educated household decisions — the same micro-decision pathway visible in every case in this table. Bangladesh is already in Table 3 as a policy over-performer (+15.8pp above its generational baseline). The connection is direct: the over-performance was the sustained expansion, and the 2014 development crossing was its result, approximately 24 years later. It is the cleanest available demonstration that income is not the mechanism.

In every case, development arrived at the generationally predicted interval after education investment — under different political systems, income levels, and provision models. Provision is not an alternative hypothesis requiring separate refutation. Every country on earth has tried provision — health campaigns, nutrition programmes, sanitation infrastructure. It is the default, not the exception. It is what educated advocates do when they disperse resources across competing priorities rather than concentrating on the fundamental cause (Section 3.4). The question was never whether provision works

independently — it is whether any country has ever developed without education. None has. Countries at comparable education levels achieve comparable life expectancy regardless of provision model. One case tests the mechanism from the opposite direction.

7.1 Cambodia: The PT Shadow

Cambodia shows what happens when education is destroyed — and how long the damage lasts. It provides the most direct available test of the PT mechanism. The country experienced severe disruption under the Khmer Rouge (1975–1979), during which the education system collapsed. The data shows two distinct stalls. The first is the exogenous shock itself. The second — a generation later — is PT's prediction.

First stall (1975–1985). Cambodia's lower secondary completion was 10.1% in 1975 and on a modest upward trajectory — like virtually every country in the region and globally during the post-1960 educational expansion. It fell to 9.4% by 1980 and was still at 9.5% in 1985 — a full decade of zero progress while the rest of the world continued to advance.

Recovery and second stall (1990–2010). After the Paris Accords and UN transitional authority (1991–1993), schools were rebuilt with sustained international investment. Completion jumped to 35.1% by 1995 on the back of that reconstruction. Then it stalled again. From 1995 to 2010, completion plateaued at 31–36%, unmoved despite continued external funding. The buildings were there; the teachers were there; the money was there. Progress did not come. External supply could recover completion *to* the pre-disruption parental baseline (~10% in 1975, amplified by international investment to ~35%); it could not push *beyond* it without the parental cohort catching up.

The first stall is the exogenous shock — the regime destroyed the education system. PT predicts the second. It is the generational shadow: the children reaching secondary school in 1996–2010 were born approximately 1982–1996 — their parents were the cohort whose own education was frozen at ~10% during 1975–1985 — the pre-disruption level, preserved in households even as the school system collapsed. Without disruption, Cambodia's trajectory (tracking neighbours like Vietnam) would have reached 25–30% by 1985. The children of the frozen cohort inherited a 10% parental baseline instead of a 25–30%

one; the plateau reflects that missing growth. Recovery from 2011 onward corresponds to the post-disruption cohort's children (parents educated after 1979) finally dominating the school-age population. Twenty-five years from the end of the disruption (1979) lands at 2004 — precisely the midpoint of the plateau.

Compare Cambodia to Vietnam over the same period: Vietnam started at 20.0% in 1960 and reached 80.8% by 2015 without interruption. Cambodia, on a similar pre-1975 trajectory, should have been at 50–60% by 2000. It was at 36%. The gap is one generation of stalled parental education, expressing itself in the educational attainment of the next.

The key evidence is the *timing* of recovery. School supply (reconstruction in 1991) predicts immediate recovery; PT predicts recovery when the parental cohort catches up (2004–2011). The buildings came back in 1991. Progress came back in 2011. That twenty-year gap is the PT shadow.

8. The Institutional Challenge

If education is the mechanism, the natural objection is: what about institutions? Acemoglu & Robinson (2012) argue that institutions — inclusive versus extractive — are the primary determinant of long-run development. Education, on this account, is endogenous: countries with inclusive institutions invest in education because their political economy rewards broad-based human capital.

The objection assumes institutional quality varies independently of education. Country fixed effects absorb time-invariant institutional characteristics, and time-varying institutional change is itself driven by education. Every measure of "institutional quality" is produced by educated populations. The civil servants, judges, journalists, health workers, and administrators who constitute institutional capacity are the educated cohort. Institutional quality does not arrive from outside the model; it is a downstream expression of the educational expansion the model measures. Controlling for it creates the same bad control problem as controlling for income (Section 4). The education-mediated account predicts that educated populations build institutions, not the reverse. The causal arrow in the institutional literature points in the wrong

direction.

The counter-cases are instructive. Qatar ($69,000 per capita, stable institutions by standard measures) delivered 3.7 percentage points below its generational education baseline in 2015. The institutions were present; the educational commitment was not; the development did not compound. The Philippines (Section 9) confirms the same point from the opposite direction.

The India–China comparison tests this directly. India has had democratic institutions, an independent judiciary, and a free press since 1947. China had none. China expanded lower secondary completion from 10% to 75% in 40 years (1950–1990) at 1.6 pp/yr — under Mao, the Cultural Revolution, and the Great Leap Forward. India expanded from 10% to 37% in the same 40 years at 0.7 pp/yr — with every institutional advantage the literature claims to matter. India's institutions did not accelerate education; China's absence of them did not prevent it. Globally, the pattern holds: among countries still expanding, mean expansion rates were 1.06 pp/yr in 1950–1975, 0.86 in 1975–2000, and 0.94 in 2000–2015. If institutional development drove educational expansion, rates should have risen as institutions matured worldwide. They did not.

The regime-type version of this objection fails on the same grounds. Korea, Taiwan, Cuba, and China were all authoritarian during their fastest educational expansion. But autocracy does not predict education speed. Merging Polity5 time-varying regime scores with the WCDE completion data (160 countries, 1950–2020, lagged 20 years to match the regime that made the schooling decision), polity2 explains less than 0.7% of the variance in education gain rates (Table A5). Democracies are slightly faster on average: 10.3 pp/decade against 8.1 for autocracies at the 20-year lag.

What autocracy produces is not speed but *variance*. 76% of autocratic countries fall below the democratic median gain rate. Autocratic skewness is $+2.4$ against $+0.6$ for democracies — a fat right tail (Korea, Cuba) and a low body (Ethiopia at 1.4 pp/decade, Chad at 1.6, Niger at 1.5). Autocracies populate both the fastest and the slowest education trajectories in the dataset. Among 50 countries that transitioned from autocracy to democracy, 30 were faster under democracy (paired mean $= +0.6$ pp/decade, $p = 0.57$). Indonesia *accelerated* after Suharto: 13.6 pp/decade under autocracy, 15.0 after 1999.

Botswana, Africa's longest continuous democracy, expanded from 8% to 84% in 45 years — comparable to autocratic Indonesia and faster than autocratic Vietnam.

Regime type is a variance amplifier, not a cause. The decision variable is education policy (Section 9), and it is orthogonal to regime type. See `scripts/regime_education_`

8.1 The Colonial Test

Acemoglu, Johnson & Robinson (2001) use settler mortality as an instrument for institutional quality: where European settlers survived, they built inclusive institutions; where they did not, they built extractive ones. The instrument cannot distinguish this from an alternative channel: where *Protestant* settlers survived, they built *schools*. The Reformation made mass literacy a theological obligation (Section 2). The Counter-Reformation removed it. Protestant colonisers (Britain, the Netherlands) transplanted mass education systems. Catholic colonisers (Spain, Portugal, France, Belgium) did not. AJR's instrument captures this difference and attributes it to institutions.

The data separates the two channels. Among 99 former colonies, education at independence (1950 lower secondary completion) explains 47% of the variance in current GDP per capita. Polity5 institutional quality explains 11%. Colonial-era education (1900 birth cohort, people schooled under colonial rule) explains 38%. Coloniser religion alone explains 7% — but adding religion to a model that already contains education raises R^2 from 0.465 to 0.466. Religion predicts GDP only because it predicts education. Once education is in the model, religion adds nothing. The channel is religion → schools → education → development, not religion → institutions → development (Table A6).

Latin America is the critical case. Spanish and Portuguese colonies were settler colonial — Europeans came and stayed in large numbers, built cities, established legal systems. AJR's framework predicts that settlement should produce inclusive institutions and therefore development. It did not. AJR require an ad hoc distinction between "inclusive" and "extractive" settler colonialism to explain why Latin America underperforms. The education account needs no such distinction: Catholic settlers brought no mass education tradition. Spain had 0.6% primary completion for its own 1875 birth cohort; Portugal

had 0.1%. They could not transplant what they did not possess. The educational base at independence was correspondingly low (mean 11% lower secondary, 1950), and development followed the education, not the institutions.

The Philippines sharpens the point further, and complicates the religious channel. Three centuries of Spanish Catholic colonialism produced almost no mass education. The American period (1898–1946) layered a public school system on top, producing 22% lower secondary completion by 1950 — comparable to Korea at 25% under Japanese colonial education. Neither country was Protestant. Both had colonial educational bases built by non-European (American, Japanese) colonial investment. Post-independence, Korea sustained the investment; the Philippines did not. The Philippines is now at 78% completion and $3,199 GDP per capita; Korea is at near-universal completion and one of the world's largest economies. The colonial education base was comparable; the post-colonial education policy diverged. Institutions without sustained education policy produce nothing.

AJR measured a downstream expression of colonial education and called it "institutions." The settler mortality instrument is, in practice, a Protestant education instrument. See `scripts/colonial_education_vs_institutions.py`.

9. Education Policy as the Decision Variable

If education is the cause, then the critical political decision is education policy — the sustained state commitment to expand education, maintained across political cycles and competing priorities. This is not a binary variable (commitment vs. no commitment) but a continuous one: expansion rate, measured in percentage points per year.

The expansion rates reported in Table 4 — from Korea's 2.14 pp/yr to India's 0.87 — show the same mechanism operating at different speeds. The speed determines whether a country crosses the development threshold in one generation or three.

Every fast-developing country in the historical record made the commitment at some point. The earliest modern example is Japan (1872): the Meiji compulsory education ordinance, implemented through repurposed temples with community teachers, no substantial budget — mandate before resources. The

pattern recurs across Korea (1953), Taiwan (1950s), and Cuba (1961) — three cases spanning the full political spectrum (Section 7). Education policy is the decision variable; regime type is not.

The colonial comparison sharpens this further. Japanese colonial education policy in Korea and Taiwan was almost singular in its depth: compulsory primary schooling enforced from the 1910s, with lower secondary infrastructure following by the 1930s. By 1950, Korea had 25% and Taiwan 18% lower secondary completion — educational bases built by a colonial state, not by household demand. The post-colonial governments in both countries then made the decisive choice: sustain and accelerate the colonial investment. American colonial education in the Philippines was similarly deep — 22% lower secondary completion by 1950, comparable to Korea. In 1960, the Philippines was one of the most prosperous countries in developing Asia: GDP per capita of $1,124 against Korea's $1,038 (constant 2017 USD), ahead of Thailand ($592), Indonesia ($598), and far ahead of India ($313) and China ($241). The Philippines had both the educational base and the income advantage. But no post-independence government sustained the educational investment. Korea made education the singular priority and expanded at 2.14 pp/yr; the Philippines drifted at roughly half that pace. The result is visible in Figure 2: comparable colonial base, comparable income, divergent post-colonial trajectory. The Philippines is the counter-case that proves education policy is a choice, not an inheritance — and that income does not make the choice for you.

None of these required high income. All required the political decision to treat education as non-negotiable. Spain is the longest counter-case: one of the wealthiest empires in history from the sixteenth century onward, it did not achieve universal lower secondary completion until the late twentieth century — roughly 450 years of wealth without mass education, and no development crossing until the wealth was finally matched by schooling.

The state's role (Section 3) is to reach children whose parents have no education — those who receive nothing from PT alone. Korea outpaced PT all the way to near-universal completion at 7–14 percentage points per five years during the main expansion (1955–1980) — fertility declining throughout.

The historical record maps onto the three speeds defined in Section 3.3: singular priority (one generation), competing priority (multi-generational), and

PT alone (century-scale).

The Korea–Costa Rica comparison is the cleanest refutation of the income-first position. Costa Rica was one of Sen's support-led cases and in 1960 had 3.5× Korea's income: $3,609 per capita against $1,038 (constant 2017 USD). The income advantage did not translate into development advantage. By 1990, Korea had grown to $9,673 — a 9-fold increase — while Costa Rica reached $6,037 — a 1.7-fold increase. Korea had the educational pace; Costa Rica had the money. Korea expanded lower secondary at 7–14 percentage points per five years (1955–1980); Costa Rica expanded at 3–6 percentage points per five years. The faster educational expansion produced the market translation that made Korea's development look "growth-mediated." Costa Rica's slower expansion produced the configuration in which state provision was the observable welfare mechanism — confirming that the "support-led" appearance is a consequence of educational tempo, not a different causal pathway.

10. Conclusion

The evidence from 187 countries over 55 years supports a single conclusion: education is the cause of human development. After removing education's contribution to GDP, income has no independent predictive power for any development outcome — not life expectancy, not fertility, not child survival, not the next generation's education (Table 2b). The apparent GDP signal was education flowing through income all along.

The institutional consequences are concrete. The aid misallocation documented in Section 1 — health over education, the SDGs ranking education behind hunger — is not a bureaucratic oversight. It is the direct expression of income-led and provision-led theory. This misallocation systematically underfunds the only mechanism that compounds across generations.

154 countries representing 80% of the world's population have already crossed — every one through education. No country has developed without it; no country with sustained educational expansion has failed to develop. The remaining 20% — concentrated in sub-Saharan Africa, with Pakistan, Afghanistan, and Yemen as the largest exceptions — is not facing an untested proposition. The question is not whether development will look like growth or welfare provision

from the outside. It is whether to make the decisive commitment: sustained educational expansion as the primary national objective, with girls' education as the highest-return lever — the household mechanism that drives fertility decline, child survival, and the next generation's schooling fires through mothers (Section 9). Cuba and Taiwan — one Soviet-aligned, one US-allied — crossed within four years of each other. Korea and Bangladesh crossed at radically different income levels. The mechanism does not care about regime type, income, or ideology. It cares whether the decision is made. Once made, the gains are irreversible — embodied in people, not stored in budgets — and visible within a single political term: completion rates rise within 5–10 years of sustained investment, providing the metric a leader needs before the generational returns arrive. The only way to remove education is to kill the people who carry it. Everything else follows from the decision to educate, transmitting through PT into every generation after.

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Development threshold data computed from World Bank WDI. USA 1960 benchmarks: TFR = 3.65, life expectancy = 69.8. Full analysis code, data, and verification outputs available at github.com/rkpagadala/education-of-nations.

A. Appendix

Table A1. Two-way fixed effects (country + year): child lower secondary completion on parental education. R^2 is the partial within-R^2 (incremental variance explained beyond country + year FEs). See `scripts/table_a1_two_way_fe.py`.

Model	Parental edu β	R^2 (within)	n	Countries
(1) FE + year: parent (< 30%)	**0.740*** **	**0.345**	790	138
(2) FE + year: parent (< 20%)	1.031***	0.351	604	119
(3) FE + year: parent (< 10%)	1.019***	0.259	359	86
(4) FE + year: parent (all)	0.080	0.009	1,683	187

*Standard errors clustered by country. *** p<0.001. Model (4) does not reach significance (p=0.08).*

Year dummies absorb the global education expansion (Section 5.3). In the pooled panel (model 4), year effects absorb nearly all the signal because the pooled panel oversamples ceiling countries where parental education has no room to move. Restricting to countries still expanding education (models 1–3),

the parental coefficient survives strongly: β=0.740 (t=5.5) at the <30% cutoff, β=1.031 (t=5.1) at <20%. The ceiling contamination that plagued the pooled two-way FE disappears when the analysis is restricted to countries where the mechanism is still operating.

Table A2. Robustness to alternative lag lengths. Across lags of 15, 20, 25, and 30 years, education coefficients retain direction and significance (p<0.001) for all three outcomes. The T+25 specification is the theoretically motivated anchor, not a cherry-picked optimum.

Table A3. Long-run panel countries (1900–2015, 28 countries). Selected on the criterion of self-determined education policy and data quality; colonial-era data for colonised countries reflects colonial investment rather than domestic policy decisions and is excluded. *East Asia:* Japan, Taiwan, South Korea. *Western Europe:* United Kingdom, France, Germany, Sweden, Norway, Denmark, Finland, Netherlands, Belgium, Switzerland, Austria, Italy, Spain, Portugal. *New World:* United States, Canada, Australia, New Zealand. *Latin America:* Argentina, Chile, Uruguay, Cuba, Costa Rica. *Other:* Sri Lanka (British Ceylon — active colonial education investment, known anomaly), Hong Kong.

Table A4. Development threshold robustness: crossing dates under three specifications (World Bank WDI data; Taiwan excluded — not in WDI). The loose spec uses generous thresholds; the main spec uses the 1960 USA values; the strict spec uses replacement fertility (TFR < 2.1) and USA life expectancy in 1972 — the year the USA itself completed the demographic transition.

Case	Loose	Main	Strict	Shift
Cuba	1971	1974	1978	7 yrs
South Korea	1984	1987	1990	6 yrs
Sri Lanka	1980	1993	2015	35 yrs
China	1990	1994	1997	7 yrs
Bangladesh	2010	2014	not yet	—

Note: The shift range is itself diagnostic. Countries with high expansion rates (Korea 2.14 pp/yr, Cuba 2.27 pp/yr, China 1.50 pp/yr) cross any reasonable

threshold definition within 6–7 years — the transition is steep enough that the choice of line barely matters. Sri Lanka's 35-year shift reflects the civil war holding life expectancy near the threshold for decades while the educational base continued accumulating beneath it; the binding constraint was disruption, not the threshold definition. Bangladesh has not yet reached replacement fertility (TFR 2.18 in 2022). Taiwan (1972, WCDE/alternative sources) is consistent with Cuba under all specifications but cannot be computed from WDI data.

Table A5. Regime type and education gain rate. Polity5 time-varying regime scores (polity2, -10 to $+10$) merged with WCDE lower secondary completion (both sexes, age 20–24). Education at age 20–24 reflects schooling 15–20 years earlier; lagged specifications attribute the education outcome to the regime that made the decision. 160 countries, 1950–2020. See `scripts/regime_education_te`

Regime lag	R^2(polity2)	Polity coef	Auto mean	Demo mean	n intervals
Concurrent (0 yr)	0.0015	+0.056	8.5	9.5	1,782
15-year lag	0.0067	+0.122	8.1	10.0	1,574
20-year lag	0.0050	+0.106	8.1	10.3	1,472

Gain rate in pp/decade. Polity coef is the OLS slope of polity2 on gain rate (positive = more democratic regimes gain faster). R^2 is the share of gain-rate variance explained by polity2 alone. Auto = polity2 ≤ -6; Demo = polity2 $\geq +6$.

Distribution of autocratic outcomes (start $< 60\%$, concurrent regime): 76% of autocratic country-intervals fall below the democratic median gain rate. Autocratic skewness $= +2.4$ vs. democratic $+0.6$: higher variance, not higher level. Among 50 countries with autocracy-to-democracy transitions, paired mean difference $= +0.6$ pp/decade (democracy faster), $p = 0.57$ (n.s.). Regime type is a variance amplifier, not a predictor of education speed.

Table A6. Colonial education vs. institutions as predictors of current GDP. 99 former colonies classified by coloniser identity and religion (Protestant: Britain, Netherlands; Catholic: Spain, Portugal, France, Belgium, Italy). Education data from WCDE v3; GDP from World Bank WDI (constant 2017 USD). See `scripts/colonial_education_vs_institutions.py`.

Predictor	R^2 (log GDP 2020)
Current education (2020 lower sec)	0.696
Education at independence (1950 lower sec)	0.465
Colonial-era education (1900 cohort primary)	0.377
Polity2 2015 (institutional quality)	0.114
Coloniser religion (Protestant = 1)	0.067
Education 1950 + coloniser religion	0.466

Each row is a separate bivariate OLS regression on log GDP per capita 2020, except the final row which includes both regressors. Adding coloniser religion to education raises R^2 from 0.465 to 0.466: religion predicts GDP only through its effect on education.

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